Effects of Tungsten on Corrosion and Kinetics of Sigma Phase Formation of 25% Chromium Duplex Stainless Steels

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ABSTRACT

Duplex stainless steels (DSS) of 25% Cr-7% Ni-x% Mo-y% W-0.25% N (x = 1.5 to 3, and y = 0 to 3) were designed with the same pitting resistance equivalent (PRE) value of 42 by varying the contents of Mo and W. Effects of W on pitting and stress corrosion of the solution-annealed or aged alloys in chlorides were examined using anodic polarization testing, critical pitting temperature measurements, and slow strain rate testing. Influences of W on degradation of mechanical properties of the alloys caused by precipitation of secondary phases during aging at 850°C were investigated using the Charpy impact and tensile tests, x-ray diffraction, and backscattered scanning electron microscopy. Resistance to pitting and stress corrosion increased with the ratio of W to Mo content. During aging, the alloys were embrittled rapidly by precipitation of the sigma (σ) phase, with the rate of embrittlement delayed significantly by increases in W content. Thus, the alloy containing 3% W-1.5% Mo exhibited the highest resistance to pitting and stress corrosion in the solution-annealed condition and the highest resistance to embrittlement induced by aging. The degree of degradation in corrosion and mechanical properties of the alloys during aging was associated closely with the amount of σ precipitates. Addition of W to 25% Cr DSS retarded nucleation and growth of the σ phase during aging, thereby delaying degradation of the corrosion and mechanical properties of the alloys. Retardation of precipitation of the σ phase by W during aging resulted from its inherently slower diffusion rate compared to that of Mo. W in the alloys caused preferential precipitation of the chi (χ) phase along the grain boundaries and, hence, inhibited nucleation and growth of the σ phase by depleting W and Mo around the χ precipitates. This beneficial effect of W on retardation of σ phase precipitation was most dominant in the alloy containing 3% W and 1.5% Mo.

KEY WORDS: aging, Charpy impact test, chi phase, backscattered scanning electron microscopy, duplex stainless steels, embrittlement, molybdenum, pitting resistance equivalent, precipitation, sigma phase, slow strain rate testing, stress corrosion cracking, tungsten, x-ray diffraction

INTRODUCTION

Austenite-ferrite duplex stainless steels (DSS) are very attractive as structural materials in energy and environmental systems, where both high mechanical strength and excellent resistance to localized and stress corrosion are required. Generally, these alloys have two to three times higher yield strength and exhibit greater resistance to localized and stress corrosion in chlorides than type 300-series austenitic SS at comparable cost.

Commercial DSS usually contain 22% to 25% Cr, 5% to 7% Ni, 3% to 4% Mo, and 0.15% to 0.35% N (wt%). Among these alloying elements, Mo has been found very effective in improving resistance to pitting and stress corrosion in chloride environments.[1-2] Thus, recent trends for new DSS are toward increasing Mo content to improve resistance to localized and stress corrosion. However, the increase of Mo content in DSS has been reported to promote precipitation of secondary phases such as the sigma (σ) and chi (χ) phases at temperatures from 600°C to 900°C.[3-4] The σ phase is a brittle, body-centered, tetragonal inter-

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TABLE 1
Compositions of the Designed DSS (%)

	Cr	Ni	Mo	W	N	Fe
3% Mo	25.26	6.39	2.72	—	0.232	Bal.
3% W-1.5% Mo	25.29	6.89	1.55	2.86	0.260	Bal.
2% W-2% Mo	25.70	7.00	1.94	2.05	0.251	Bal.

metallic compound that is rich in Cr and Mo,[4-6] with a larger volume fraction than any other phases in alloys.[4,7] The χ phase also is an intermetallic compound with a manganese sulfide (MnS) structure.[5,8] Since these two phases often coexist, it is difficult to separate inherent effects of the χ phase on mechanical and corrosion properties of the alloys from those of the σ phase.

It is well known that mechanical properties of DSS can be deteriorated seriously by formation of small amounts of σ phase.[4,7] Thus, production of thick pipes or bars with large diameters are limited because of precipitation of the σ phase in the interior of products where the cooling rate is relatively slow after solution-annealing. Precipitation of σ also depletes surrounding phases of Mo and Cr, leading to a reduction in corrosion resistance of DSS.[7,9-11] Thus, it is very important to control the kinetics of σ phase formation in DSS.[1,12-13] Some results have shown that partial substitution of Mo by W increases localized corrosion resistance and significantly reduces the rate of σ phase formation in 22% Cr and 25% Cr DSS.[14-15] In spite of these beneficial effects, few attempts have been made to study the effects of W on DSS systematically.

The purpose of the present work was to clarify effects of partial substitution of Mo by W on corrosion and σ phase formation kinetics of 25% Cr DSS by observing changes in mechanical properties and phase distributions with aging treatment.

EXPERIMENTAL

Experimental alloys were designed to have the same value (42) of pitting resistance equivalent (PRE) by varying the Mo and W contents for Fe-25% Cr-7% Ni-x% Mo-y% W-0.25% N (x + 1/2, and y = 3) alloys. The PRE of DSS, a criterion for pitting resistance, is defined by Equation (1) and calculated using the chemical compositions (wt%) of major alloying elements such as Cr, Mo, W, and N that significantly affect the resistance of SS to pitting corrosion:[16-18]

$$\mathrm{PRE} = \%\mathrm{Cr} + 3.3(\%\mathrm{Mo} + 1/2\%\mathrm{W}) + k\%\mathrm{N} \qquad (1)$$

In the case of DSS, the value of k has been reported as 30.[16] According to the equation, W enhances the resistance to pitting corrosion of DSS half as much as Mo. However, a systematic study of the effects of W on resistance to pitting corrosion of DSS has not been made. Thus, the proportional constant for W in the PRE equation needs to be determined correctly through more tests.

The experimental alloys were high-purity heats melted in a laboratory-scale vacuum induction furnace and cast in 25-kg ingots. Chemical compositions of the alloys are presented in Table 1. An alloy containing 6% W and having the same PRE value of 42 was designed and melted, but it cracked so badly during hot-rolling that it was rejected in further tests. Oh, et al., reported that a 22% Cr DSS containing 6% W exhibited a much lower toughness than the alloy with 3% Mo or 3% W-1% Mo when aged at 850°C and that it did not reveal the beneficial effect of W on resistance to pitting corrosion in chloride solution.[15] In the present work, each alloy was designated as 3% Mo, 3% W-1.5% Mo, and 2% W-2% Mo after their compositions. The alloys were prepared in the form of cold-rolled sheets 3 mm thick and hot-rolled plates 12 mm thick. All these materials were solution-annealed for 2 h at 1,050°C and then aged at 850°C for 30 min, 1 h, and 10 h.

Resistance to localized corrosion was examined by measuring the pitting potential or the critical pitting temperature (CPT) of the alloys.[18-21] Anodic polarization tests were performed to determine the pitting potential of the alloys in deaerated 4 M sodium chloride (NaCl) solution at 80°C. CPT for the alloys were measured using the potentiostatic method.[21] The electrochemical cell for both tests consisted of a 1-L multineck flask, as specified in ASTM G 5,[22] with a platinum counter electrode and a saturated calomel electrode (SCE) positioned in a salt bridge with a high-silica tip. To determine the CPT, the sample was polarized from the corrosion potential to 600 mV_{SCE} at a scan rate of 1 mV/s. Then, the solution was heated at 0.5°C/min while simultaneously measuring the change in anodic current of the sample polarized at 600 mV_{SCE}. CPT was measured to be that at which the anodic current increased abruptly. CPT was measured at least four times for the same alloy to certify reproducibility of the result.

Stress corrosion cracking (SCC) tests were conducted using slow strain rate testing (SSRT) at a rate of 6 x 10^{-6} /s in boiling 35% and 42% magnesium chloride ($MgCl_2$) solutions. The solutions were deaerated with nitrogen throughout the tests. Tensile

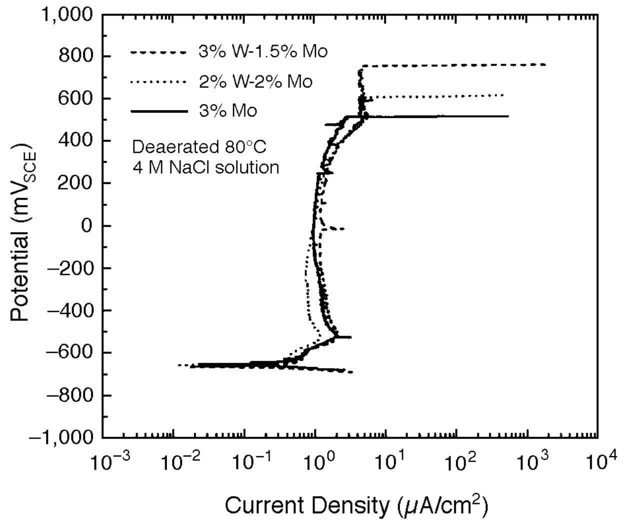


FIGURE 1. *Effects of W substitution for Mo on anodic polarization responses in deaerated, 80°C, 4 M NaCl solution.*

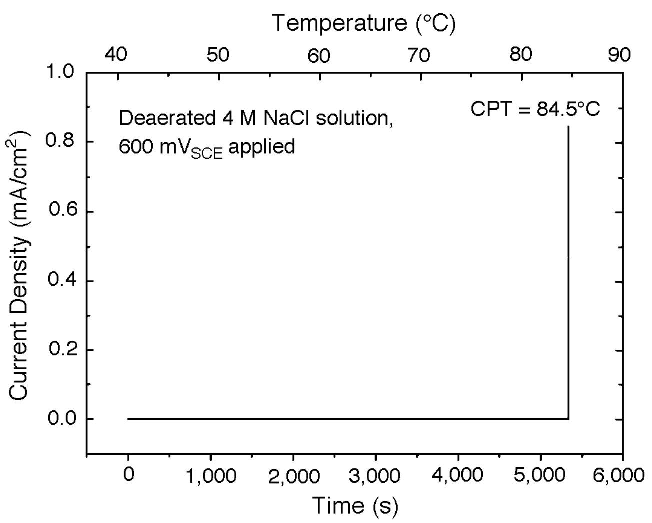


FIGURE 2. *CPT of solution-annealed 3% Mo polarized to 600 mV$_{SCE}$ in deaerated, 4 M NaCl solution.*

specimens used for SSRT were cut parallel to the rolling direction of the sheets with a gauge section 20 mm long and 2.5 mm wide.

Mechanical tests such as the Charpy V-notch impact test and tensile tests were conducted at room temperature to compare the effect of aging on degradation of mechanical properties of the alloys. Standard impact test specimens were machined from the plate 12 mm thick.[23] Tensile specimens were cut parallel to the rolling direction of the sheets 3 mm thick with a gauge section 25 mm long and 6.35 mm wide.

Precipitation of secondary phases such as σ and χ for aged specimens was detected by x-ray diffraction (XRD) using Cu K_α radiation with a monochromator and characterized by back-scattered scanning electron microscopy (SEM). While chemical compositions of the coarse secondary precipitates were measured using energy-dispersive x-ray (EDX) analysis, those of the fine ones were analyzed by transmission electron microscopy (TEM)-EDX. Thin foil specimens for TEM analysis were prepared electrochemically using 5% perchloric acid ($HClO_4$) in acetic acid (CH_3COOH).

RESULTS AND DISCUSSION

Effects of W on Pitting and SCC Resistance

Figure 1 shows typical anodic polarization responses for solution-annealed 3% Mo, 3% W-1.5% Mo, and 2% W-2% Mo alloys in deaerated 4 M NaCl solution at 80°C. All the alloys had corrosion potentials of ~ −630 mV$_{SCE}$ and exhibited a passive state in anodic polarization from the corrosion potentials. The pitting potentials for 3% Mo, 3% W-1.5% Mo, and 2% W-2% Mo were 609 mV$_{SCE}$, 758 mV$_{SCE}$, and 515 mV$_{SCE}$, respectively. These values were close to the averages determined from five polarization tests for each alloy. The 3% W-1.5% Mo alloy, however, exhibited the highest pitting potentials. These results showed the resistance to localized corrosion of DSS containing Mo and/or W was maximum when they were alloyed by 3% W and 1.5% Mo, even if the PRE values of the alloys were equal. Similar results have been obtained by other researchers. Oh, et al., showed that among 22% Cr DSS (22% Cr-5.5% Ni-0.15% N) containing 3% Mo, 3% W, 6% W, and 3% W-1% Mo, respectively, the alloy containing 3% W-1% Mo revealed the highest pitting potential in deaerated 3.5% NaCl solution at 80°C.[15]

Since the pitting potential of DSS measured in every anodic polarization test was scattered in the range from 30 mV$_{SCE}$ to 100 mV$_{SCE}$ from the value of pitting potential shown in Figure 1, the resistance to pitting corrosion of the experimental alloys was reconfirmed by measuring the CPT. Whereas the varying parameter in an anodic polarization test is applied potential, that in a CPT test is solution temperature. Actually, CPT is preferred as a parameter in evaluating the resistance to pitting corrosion of DSS over the pitting potential.[18-21] Figure 2 shows a typical current-vs-time curve for the 3% Mo alloy obtained in a CPT measurement test. During the test, the sample was kept at a passivation potential of 600 mV$_{SCE}$ in deaerated 4 M NaCl solution, with the solution being heated at a rate of 0.5°C/min from 40°C. It was evident that the 3% Mo alloy maintained its passivity without localized film breakdown below a solution temperature of 84.5°C, at which the anodic current increased abruptly, indicating pitting corrosion occurred. The solution temperature of 84.5°C at which the anodic current density abruptly increased was measured to be the CPT for the 3% Mo alloy. Similar results on the CPT were obtained for other alloys. By observing the surfaces of the specimens after CPT tests, only one conspicuous pit was found to be formed.

Figure 3 shows CPT values of solution-annealed samples for the 3% Mo, 2% W-2% Mo, and 3% W-1.5% Mo alloys. Average CPT for the 2% W-2% Mo and 3% W-1.5% Mo alloys were measured to be 88.5°C and 91°C, respectively, which showed that resistance to localized corrosion increased in the order of 3% Mo, 2% W-Mo, and 3% W-1.5% Mo (i.e., with the ratio of W to Mo content even if the PRE values for the alloys were equivalent). Results in Figure 3 corresponded to those of pitting potentials presented in Figure 1.

To examine the effect of W addition on SCC resistance, SSRT was conducted for each alloy in the solution-annealed condition. For all the alloys, SCC did not occur in deaerated boiling 35% $MgCl_2$ solution, indicating all the experimental alloys were highly resistant to SCC in chlorides. However, all the alloys were susceptible to cracking in boiling 42% $MgCl_2$ solution (Figure 4). The 3% W-1.5% Mo alloy revealed the highest resistance to SCC when evaluated in terms of either the strain to failure or the maximum stress in stress-strain curves as shown in Figure 4. SSRT results suggested the SCC resistance of 25% Cr DSS increased with the ratio of W to Mo content. Resistance to SCC was shown to parallel that for pitting corrosion, showing a close relationship in resistance to SCC and pitting corrosion.

Effects of Aging on the Mechanical Properties

Results of the impact tests for 3% Mo, 3% W-1.5% Mo, and 2% W-2% Mo alloys that were aged at 850°C are shown in Figure 5. In the solution-annealed condition, the 3% Mo, 3% W-1.5% Mo, and 2% W-2% Mo alloys revealed impact toughness values of 272 J, 284 J, and 262 J, respectively. Charpy impact values for the three alloys reduced significantly with aging, probably as a result of precipitation of secondary phases. The precipitation of σ phase, for example, is known to be very detrimental to the toughness of SS containing high contents of Cr and Mo.[13] Among the designed alloys, the 3% W-1.5% Mo alloy exhibited the highest resistance to embrittlement induced by aging. For the alloys aged for 30 min, the 3% W-1.5% Mo alloy showed comparably high impact toughness of 103 J, but the other alloys revealed toughness as low as ~ 32 J. When aged for over 1 h, the toughness of all the alloys fell to very low values, implying that amounts of σ precipitate in all the alloys were so large that the decrease in toughness leveled out.

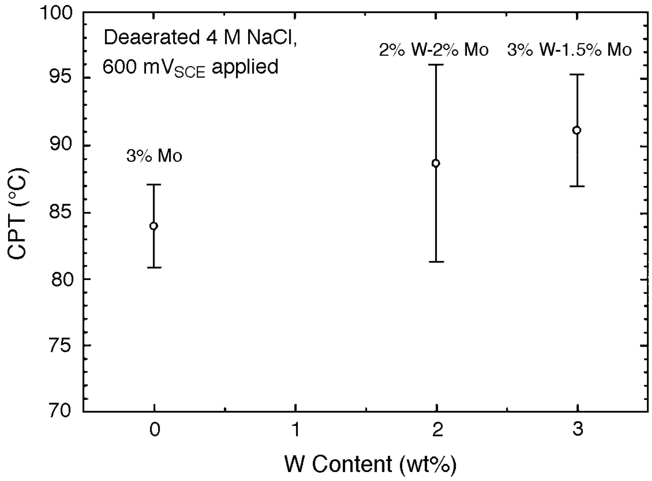


FIGURE 3. *Effect of W on CPT of 25% Cr DSS polarized to 600 mV$_{SCE}$ in deaerated, 4 M NaCl solution.*

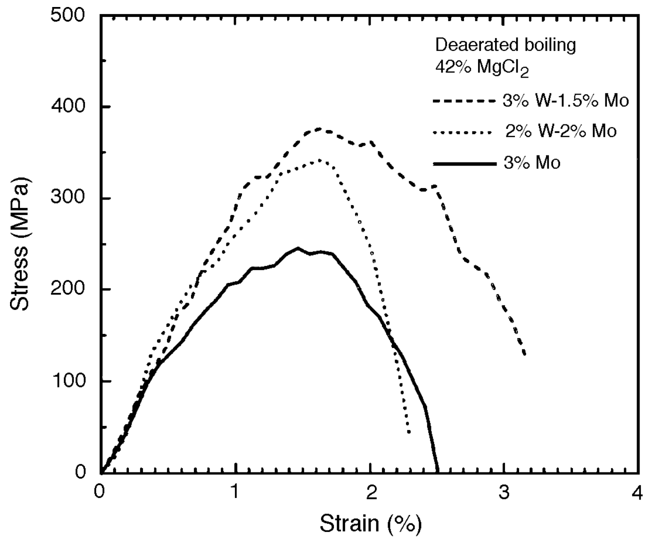


FIGURE 4. *Stress-vs-strain curves for solution-annealed alloys of 3% Mo, 3% W-1.5% Mo, and 2% W-2% Mo in deaerated boiling 42% $MgCl_2$. Specimens were strained at 10^{-6} /s by SSRT.*

Nilsson and Wilson reported that the impact energy of SAF 2507[†] (UNS 32750)[(1)] was only 16 J after aging for 10 min at 850°C.[7] Golovanenko, et al., also suggested that the impact energy of UNS 32550, which contains 25% Cr and 3% Mo, was as low as 10 J when aged for 30 min at 800°C to 900°C.[24] These results showed the degradation rate of the toughness is quite fast in 25% Cr-based DSS containing 3% to 4% Mo without W. In contrast, Oh, et al., suggested that 22% Cr DSS containing 3% W or 3% W-1% Mo had higher toughness than that containing 3% Mo when aged for 30 min at 850°C.[15] According to their results, however, the alloy containing 6% W, having the same PRE as that containing 3% Mo, showed no better resistance to embrittlement on aging at 850°C than the alloy containing 3% Mo. This implied that

[†] Trade name.
[(1)] UNS numbers are listed in *Metals and Alloys in the Unified Numbering System*, published by the Society of Automotive Engineers (SAE) and cosponsored by ASTM.

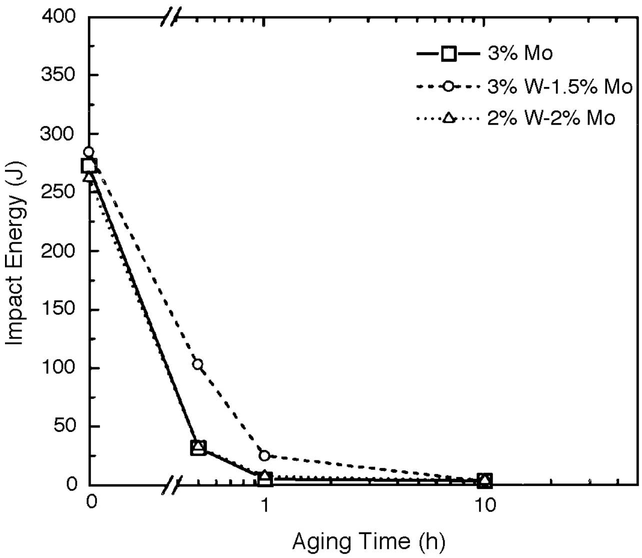


FIGURE 5. *Effects of aging at 850°C on the Charpy impact toughness of the designed alloys: 3% Mo, 3% W-1.5% Mo, and 2% W-2% Mo.*

the degradation in toughness of DSS with aging is delayed by the partial substitution of Mo by W.

Figure 6 represents the effects of aging at 850°C on the tensile properties of 3% Mo, 3% W-1.5% Mo, and 2% W-2% Mo alloys. In the solution-annealed condition, these alloys showed elongations of 39.8%, 36.0%, and 40.8%, respectively, and yield strengths of 619 MPa, 610 MPa, and 594 MPa, respectively. These values of elongation and yield strength were fairly high in comparison with those of commercial duplex alloys, primarily because of the high purity of the experimental alloys used in this study. As Figure 6 shows, all the alloys were getting brittle with aging, as confirmed by a gradual decrease in elongation and an increase in Young's modulus. Degradation in elongation of the alloys with aging is summarized in Table 2. As was the case of toughness, reductions in ductility were different depending upon the alloy composition, especially the ratio of W to Mo in the alloys. As the ratio of W to Mo of the alloys increased (3% Mo > 2% W-2% Mo > 3% W-1.5% Mo), the reduction in elongation of the alloys with aging decreased (Table 2). Thus, 3% Mo and 2% W-2% Mo were degraded so fast that the elongations fell below 15% simply by 1-h aging, whereas 3% W-1.5% Mo maintained a relatively high elongation of 29.2% in the same aging condition.

Effects of W Addition on Growth of σ Phase

The gradual degradation of mechanical properties of the experimental alloys with aging appeared to be related closely to precipitation of secondary phases such as σ and χ. Furthermore, differences in the degradation rate of toughness and tensile properties among the three alloys were thought to be associated with the influence of W on growth rates of

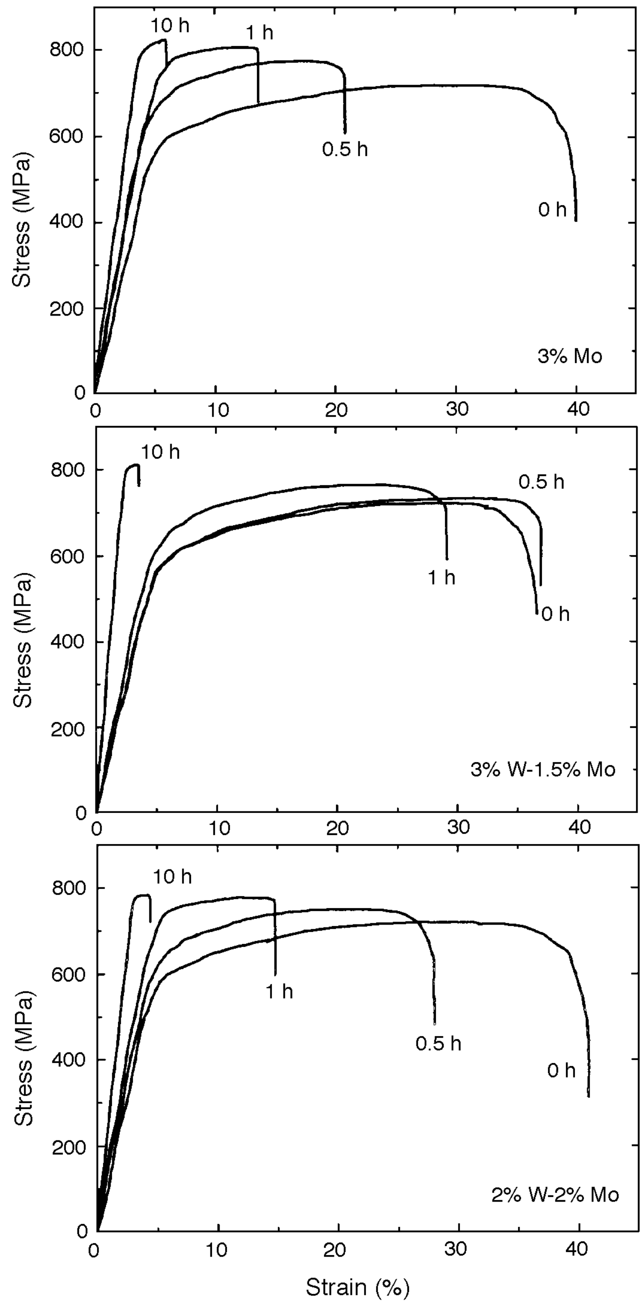


FIGURE 6. *Effects of aging at 850°C on stress-vs-strain behavior of the designed alloys: 3% Mo, 3% W-1.5% Mo, and 2% W-2% Mo.*

the secondary phases. Figure 7 shows the XRD patterns of the alloys subjected to aging for 1 h at 850°C in which the diffraction peaks for α, γ, and σ phases were observed. For 3% Mo and 2% W-2% Mo alloys, the diffraction peaks for σ phase appeared distinct around 42° of 2θ and weak between 45° to 50° of 2θ, whereas no detectable σ peaks appeared in 3% W-1.5% Mo alloy, presumably as a result of the relatively lower volume ratio of the σ phase compared with those of the former two alloys. This was confirmed by SEM micrographs of the σ phase in the alloys (Figure 8). For the alloys subjected to aging for

1 h at 850°C, the amount of σ phase appeared to decrease in the order of 3% Mo, 2% W-2% Mo, and 3% W-1.5% Mo alloys. By relating the degradation in toughness and ductility of the alloys induced by aging (Figures 5 and 6) to both the amount of the σ precipitates (Figure 8) and the W content, it was concluded that the degree of embrittlement of the alloys during aging was proportional to the amount of σ precipitates that significantly decreased when alloyed with 3% W-1.5% Mo.

Except for the σ phase, no other secondary phases were detected in the XRD intensity graphs, even after aging for 10 h. This may have been a result of the small volume fraction of minor phases.

To observe the effects of W on secondary phase formation, polished samples of aged material were investigated using back-scattered SEM. Figure 9 represents back-scattered SEM images for each alloy aged for 10 min at 850°C. For the 3% Mo alloy shown in Figure 9(a), a small portion of σ precipitates was observed. This appeared to be white because of the high content of heavy Mo. Most σ precipitates nucleated at grain-boundary triple points, so their shapes were multiangular types along grain boundaries. The α phase appeared darkest, and the γ phase brighter than α but still dark in the back-scattered images. In contrast, for the 3% W-1.5% Mo shown in Figure 9(b), small particles brighter than the σ precipitates were precipitated predominantly along grain boundaries with some σ phases. The brighter precipitates were χ phases richer in Mo and W than the σ phase. Many authors have observed the precipitation of χ phase in DSS without W, such as in UNS 32750.[4,7,25] However, no evidence of precipitation of the χ phase was found in the 3% Mo alloy that was equivalent to UNS 32750. The possibility still remained that the χ phase of 3% Mo had vanished in shorter aging than 10 min. To confirm this, the 3% Mo alloy aged for 5 min was observed by back-scattered SEM. However, no χ precipitates were observed.

Figure 9 shows that the precipitates of secondary phases in the 3% W-1.5% Mo alloy appeared finer in size but larger in number compared to those in the 3% Mo alloy. However, the total volume of secondary phases in the 3% Mo alloy was greater compared to that in the 3% W-1.5% Mo.

Figure 10 shows the back-scattered SEM images of 3% Mo and 3% W-1.5% Mo after aging for 1 h at 850°C. As was the case of 10-min aging, the χ precipitates were not observed in the 3% Mo alloy. In contrast, small χ particles precipitated in the 3% W-1.5% Mo alloy. Further, all the α/γ boundaries were densely covered with coarse σ precipitates in the 3% Mo, whereas for the 3% W-1.5% Mo alloy, a large portion of the α/γ boundaries were free from the σ phase, and some of the boundaries contained only χ precipitates. Evidently, the χ precipitates remained small even after 1 h aging at 850°C, whereas the

TABLE 2

Elongation (%) of 3% Mo, 3% W-1.5% Mo, and 2% W-2% Mo with Aging at 850°C

	Aging Time (h)			
	0	**0.5**	**1**	**10**
3% Mo	39.8	23.6	13.6	5.0
3% W-1.5% Mo	36.0	37.0	29.2	3.6
2% W-2% Mo	40.8	28.0	14.8	4.4

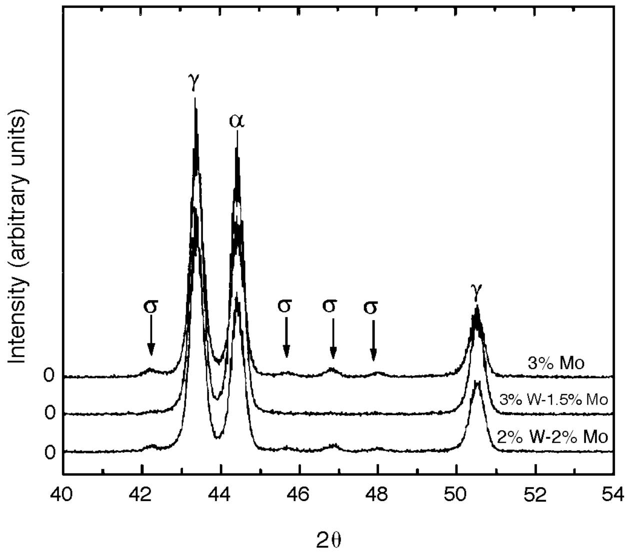


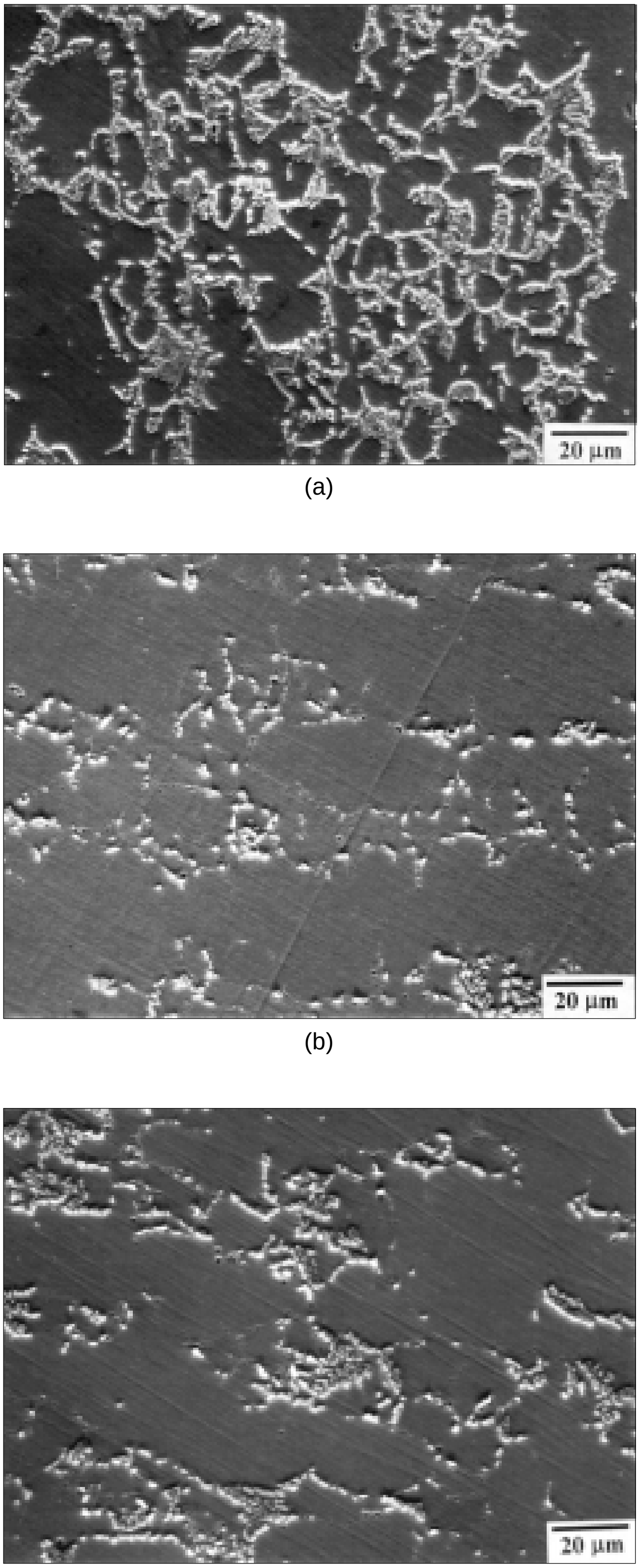
FIGURE 7. *XRD patterns, obtained by Cu K_α radiation with a monochromator, of the designed alloys aged for 1 h at 850°C and showing the effects of W substitution for Mo on growth of the σ phase.*

σ precipitates had grown so large that a large portion of ferrite phase was transformed to σ.

Figures 8 through 10 clearly demonstrate that the growth rate of σ phase in 3% W-1.5% Mo was slower than in 3% Mo or 2% W-2% Mo during aging. Thus, it could be suggested that there exists a specific alloying ratio of W to Mo at which the growth of σ phase in DSS can be delayed significantly. When the effects of aging on precipitation of σ phase was compared to those on the degradation in mechanical properties for the alloys, it was concluded that the degree of embrittlement of the alloys during aging was proportional to the growth of σ phase.

Role of W in Reducing σ Phase Growth Rates

Questions remained as to why the χ phase precipitated only in the alloys containing W, and why the growth of σ phase was delayed significantly in the 3% W-1.5% Mo alloy. To solve these questions, the compositions of each phase formed in the 3% W-1.5% Mo aged for 10 min at 850°C were measured via TEM-EDX and SEM-EDX (Table 3). The σ and χ phases were much richer in Mo and W contents than the α or γ. The χ phase especially contained very high



(a)

(b)

(c)

FIGURE 8. *SEM micrographs showing σ phase precipitated during aging at 850°C for 1 h: (a) 3% Mo, (b) 3% W-1.5% Mo, and (c) 2% W-2% Mo. Each specimen was etched electrochemically in 10 g potassium hydroxide (KOH) + 100 mL water (H_2O) under 5-V direct current for ~ 30 s.*

contents of W (22 wt%) compared to 6.27 wt% in the σ phase. However, there was no large difference in Mo content between the two phases (3.56% and 3.07%, respectively, for the χ and σ phases). Thus, during aging, W was likely to segregate preferentially in χ rather than in σ. In other words, W had to be a strong stabilizer for the χ phase. This answered the question as to why the χ phase precipitated only in DSS containing W, such as 3% W-1.5% Mo and 2% W-2% Mo.

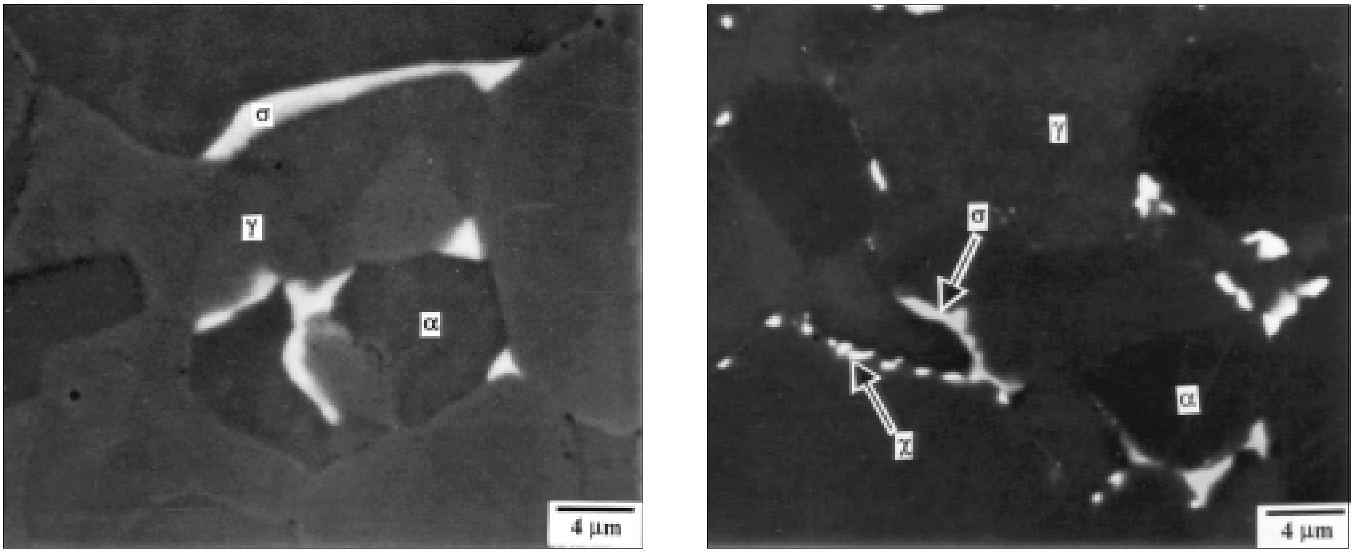
Since formation of σ phase in the 3% W-1.5% Mo alloy required high concentrations of Mo and/or W, the preferential precipitation of χ in the alloy during the initial period of aging would inhibit the nucleation and growth of σ phase by depleting W and Mo adjacent to the χ precipitates. This was confirmed in Figure 9(b), where a small number of σ precipitates were formed with a large number of χ precipitates along grain boundaries, and the σ precipitates were formed in grain boundaries where the χ precipitates had not been formed. Thus, preferential precipitation of χ phase in 3% W-1.5% Mo alloy during the initial period of aging was one reason for the retardation in the growth of σ phase in 3% W-1.5% Mo compared to that in 3% Mo alloy. The other reason for the retardation in the nucleation and growth of the σ phase in the 3% W-1.5% Mo alloy may have been the inherent difference in diffusion rate between W and Mo. It has been reported that the diffusion rate of W at 850°C was 10 to 100 times slower than that of Mo in iron or ferrous alloys.[26-28]

The χ phase of 3% W-1.5% Mo contained a relatively high W content (Table 3). Thus, the formation of fine χ precipitates as shown in Figure 10 resulted primarily from the difficulty in forming the χ precipitate that required a large amount of W to be diffused at a slow rate. Since W should have been supplied for the growth of the σ phase in 3% W-1.5% Mo, it would take more time for the σ phase to grow in 3% W-1.5% Mo alloy compared to the case in the 3% Mo alloy in which the σ phase contained only Mo, a fast diffusing element.

Consequently, W contributed to retarding the nucleation and growth of σ phase in these DSS. This effect of W appeared most dominant in the 3% W-1.5% Mo alloy that contained more W and less Mo than the other alloys.

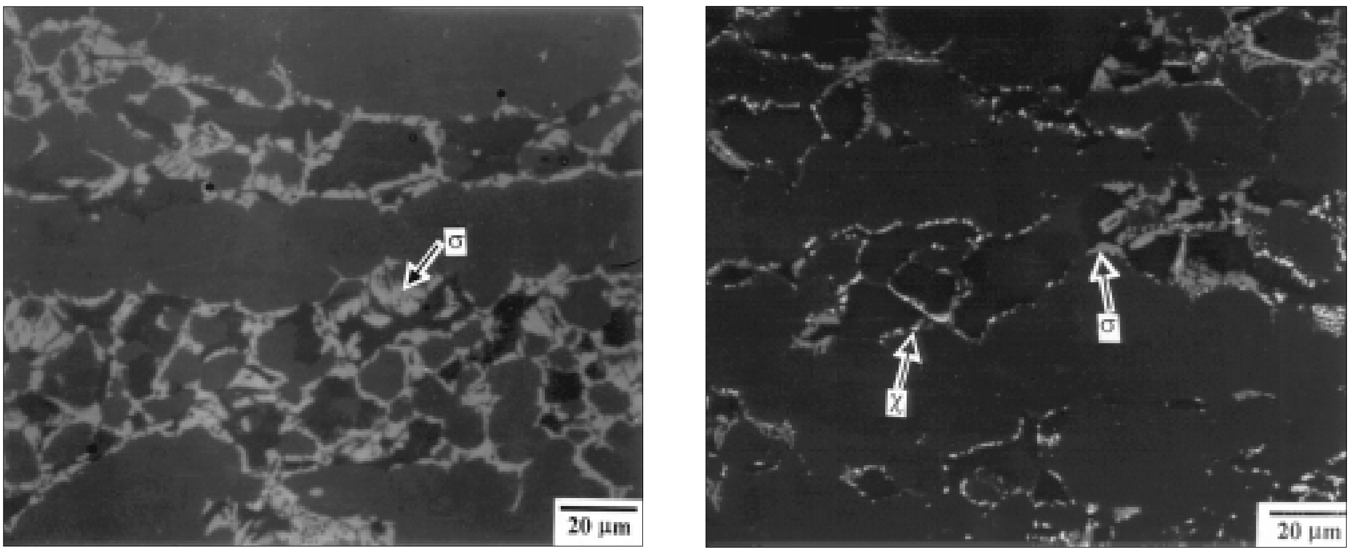
Effects of Aging on Pitting Corrosion Resistance

Effects of aging at 850°C on CPT of 3% Mo and 3% W-1.5% Mo alloys are presented in Figure 11. CPT of the alloys decreased with aging, with the CPT of 3% W-1.5% Mo being higher than that of 3% Mo for the same aging condition. For the 3% Mo alloy, the average CPT decreased gradually with aging, at 84°C for the solution-annealed material and 69°C for the material aged for 600 min. In contrast, CPT of



(a) (b)

FIGURE 9. *Back-scattered SEM micrographs showing the phases in: (a) 3% Mo and (b) 3% W-1.5% Mo alloys, both aged for 10 min at 850°C.*



(a) (b)

FIGURE 10. *Back-scattered SEM micrographs showing the precipitates of secondary phases (σ and χ) for: (a) 3% Mo and (b) 3% W-1.5% Mo that were subjected to aging for 1 h at 850°C. As in the 10-min aging condition, χ precipitates were observed only in 3% W-1.5% Mo.*

3% W-1.5% Mo aged for 10 min was nearly the same as that for the solution-annealed specimen, 90°C in average for the material aged for 10 min. However, the 3% W-1.5% Mo alloy, when aged for > 30 min at 850°C, exhibited a CPT value 10°C to 15°C lower than that for the solution-annealed. Therefore, it was concluded that addition of W to 25% Cr DSS improved pitting corrosion resistance of the alloys.

Reduction in CPT of the alloys with aging resulted from precipitation of σ and χ phases and the resultant depletion of Cr and Mo/W adjacent to the precipitates. According to Wilms, et al., who studied 24.01% Cr-7.11% Ni-3.89% Mo-0.29% N DSS, the resistance to pitting and crevice corrosion resistance of that alloy in seawater was reduced on aging for > 7 min.[11] They also reported that initiation of localized corrosion took place around the σ phase as a result of depletion of Cr and Mo, while the σ phase itself appeared unaffected. This was confirmed in the present study by SEM images of the pits shown in Figure 12. Evidently, pits were formed in sites where the σ phase precipitated densely. The fine precipi-

TABLE 3
Compositions (wt%) of α, γ, σ, and χ Phases Measured Using TEM-EDX for 3% W-1.5% Mo Aged at 850°C for 10 min

	Fe	Cr	Ni	Mo	W
Ferrite (α)	63.15	27.00	4.94	1.72	3.19
Austenite (γ)	64.73	22.18	9.37	1.31	2.41
Sigma (σ)[(A)]	58.43	27.91	4.33	3.07	6.27
Chi (χ)	50.05	19.40	4.44	3.56	22.54

[(A)] Measured by SEM-EDX.

tates of the σ phase were observed to be unaffected, whereas dissolution within pits occurred between the fine precipitates where Cr and Mo/W probably were depleted.

While mechanical properties such as toughness and ductility for the aged alloys were reduced significantly by precipitation of the σ phase (Figures 5 and 6), resistance to pitting corrosion of the aged alloys was relatively high, as confirmed by the high CPT for the aged alloys that were just ~ 15°C lower than that for the solution-annealed, as shown in Figures 11(a) and (b). The reason for the small difference in resistance to pitting between the solution-annealed alloy and the aged one primarily was the fact that the degree of Cr depletion in areas adjacent to the σ precipitate was relatively low because of the small difference in Cr content between the ferrite phase and the σ phase. Table 4 shows that Cr content of the σ phase was only 3% to 4% higher than that of the ferrite phase (Table 3), where the σ phase was nucleated. However, a significant depletion of Mo or W would occur in areas adjacent to the precipitates of σ phase, thereby contributing to the reduction in resistance to pitting.

It was noted in Figure 11 that, whereas the CPT of the 3% Mo alloy was lowered by 7°C to 8°C by 10 min aging at 850°C, CPT of the 3% W-1.5% Mo alloy was not reduced for the same aged condition. This could be explained by comparing the size and distribution of precipitates between the two alloys (Figure 9). Coarse σ phase was precipitated in the 3% Mo alloy, whereas very fine σ and χ phases were precipitated in the 3% W-1.5% Mo alloy. Thus, it appeared that the reduction in resistance to pitting corrosion of the alloys during aging occurred when precipitates of σ phase grew to some degree. Further, the higher resistance of the 3% W-1.5% Mo alloy to reduction in pitting resistance during aging in comparison with that of the 3% Mo alloy resulted from the retarding effect of W on nucleation and growth of the σ phase.

CONCLUSIONS

❖ For the designed alloys, resistance to pitting and stress corrosion in chlorides increased with the alloying ratio of W to Mo. Hence, the 3% W-1.5% Mo alloy exhibited the highest resistance to pitting and stress corrosion.

❖ All the designed alloys showed excellent toughness and tensile properties in the solution-annealed condition. However, the alloys were embrittled rapidly with aging at 850°C, with the rate of embrittlement delayed significantly with increases in W content. Thus, the 3% W-1.5% Mo alloy revealed the highest resistance to embrittlement induced by aging and maintained a high Charpy impact value of 103 J without an appreciable loss in ductility after aging.

❖ W in the designed alloys appeared to be a strong stabilizer for the χ phase that contained high contents of W and Mo. The preferential precipitation of

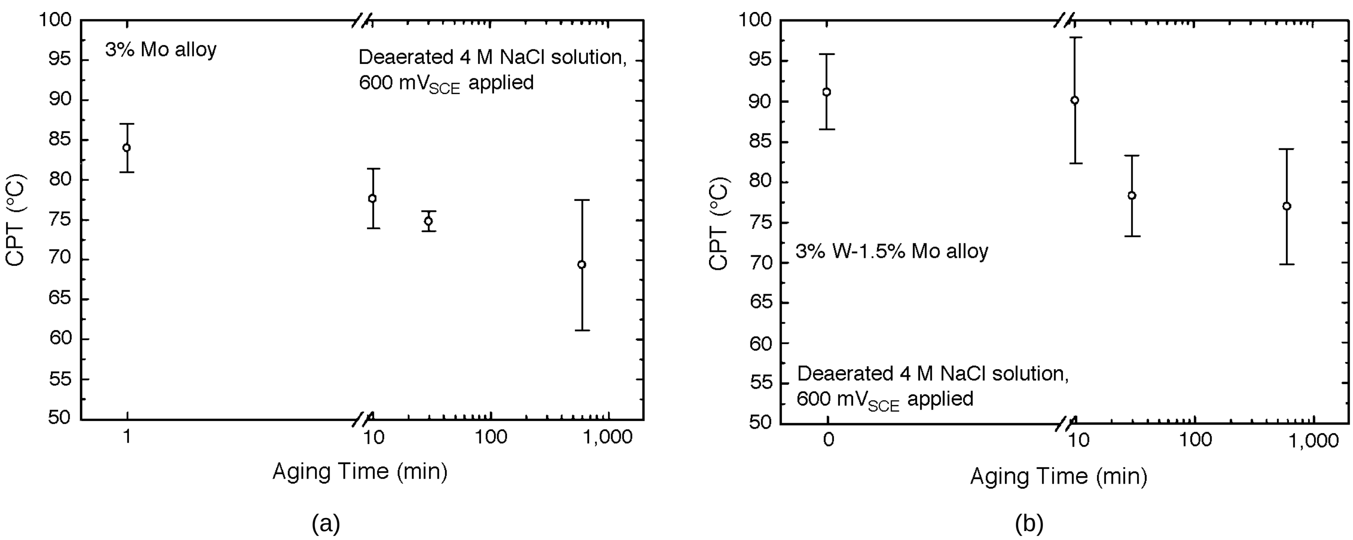


FIGURE 11. *Effect of aging at 850°C on resistance to CPT of: (a) 3% Mo and (b) 3% W-1.5% Mo alloy.*

the χ phase in the alloy containing W during the initial period of aging contributed to retarding nucleation and growth of the σ phase by depleting W and Mo along grain boundaries. Since the σ phase in the 3% W-1.5% Mo alloy contained a relatively high content of W, the inherently low diffusivity of W compared to that of Mo in the alloy may have contributed to the delay in nucleation and growth of the σ phase during aging. This effect of W appeared most dominant in the 3% W-1.5% Mo alloy.

❖ Degradation in corrosion and mechanical properties of the alloys during aging was associated closely with the amount of σ precipitates. With increases in the ratio of W to Mo in the alloys, nucleation and growth of the σ phase was retarded significantly, thereby delaying the degradation in corrosion and mechanical properties of the alloys during the initial period of aging.

❖ For the aged alloys, pits were formed in sites where the σ phase precipitated densely. Dissolution within a pit occurred between the fine σ precipitates where Cr and Mo/W were depleted.

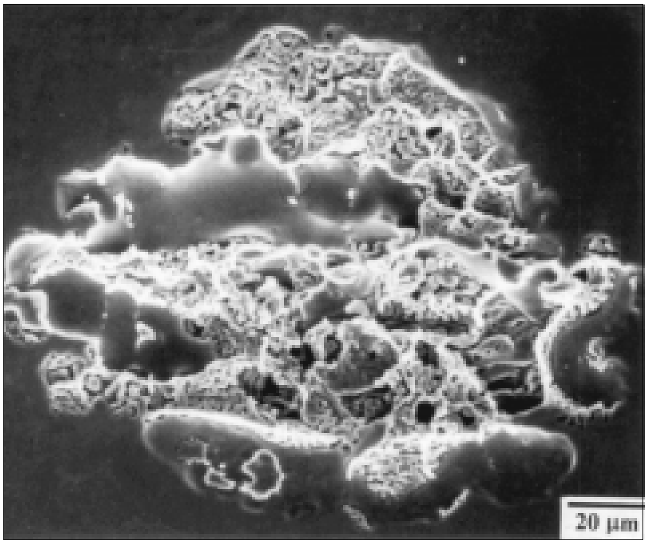


FIGURE 12. *Pit morphology of the 3% Mo alloy aged for 10 h at 850°C after CPT measurement testing in deaerated 4 M NaCl solution under 600 mV_{SCE} anodic polarization.*

TABLE 4
Compositions (wt%) of σ Phase Measured Using SEM-EDX for 3% Mo and 3% W-1.5% Mo Aged at 850°C for 1 h

	Cr	Ni	Mo	W	Fe
σ in 3% Mo	30.82	3.03	9.72	–	Bal.
σ in 3% W-1.5% Mo	30.10	3.39	7.12	6.57	Bal.

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